

Auteur: Marie Van Nuffel
Studierichting: Informatica
Oefeningenreeks 1B nr. 43.5

$$z^2 - 7z + 13 + i = 0$$

$$D = (-7)^2 - 4 \cdot (13 + i)$$

$$= 49 - 52 - 4i$$

$$= -3 - 4i$$

$$-3 - 4i = (x + yi)^2 = x^2 + 2xi - y^2$$

Stelsel oplossen:

$$\Leftrightarrow \begin{cases} x^2 - y^2 = -3 \\ 2xy = -4 \end{cases}$$
$$\Leftrightarrow \begin{cases} x^2 - y^2 = -3 \\ x^2(-y)^2 = -\frac{1}{4} * (-4)^2 \end{cases}$$

Pas de vergelijking toe:

$$\lambda^2 + 3\lambda - 4 = 0$$

$$\lambda_1 = \frac{-3 + \sqrt{9 + 16}}{2} = 1 \text{ en } \lambda_2 = \frac{-3 - \sqrt{9 + 16}}{2} = -4$$

$$\Rightarrow x^2 = 1 \text{ en } y^2 = -4$$

$$k = 1 + 2i \text{ en } -k = -1 - 2i$$

$$z_1 = \frac{7 + 1 + 2i}{2} = 4 + i$$

$$z_2 = \frac{7 - 1 - 2i}{2} = 3 - i$$

References